VedShare: Development of AI-Powered Study Management System with Resource Marketplace.

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DOMAIN: - SOFTWARE ENGINEERING SYSTEM, (QUALITY EDUCATION , TRL-5)

GUIDE: - MRS. LEENA CHAKRABORTY

INTRODUCTION

1. Brief Background of the Project Domain

1. Software Engineering Systems: Focused on building reliable, scalable, and user-centric applications.
2. Educational Technology (EdTech): Leveraging software to enhance accessibility, learning efficiency, and collaboration.
3. Quality Education (SDG Goal 4): Ensures inclusive, equitable, and affordable access to academic resources.
4. Technology Readiness Level (TRL-5): Concept validated in relevant environment with partial implementation and testing.

PROBLEM STATEMENT

Students in higher education struggle to **access, organize, and manage academic resources** efficiently.
Study materials (notes, books, PDFs) are **scattered across multiple unconnected platforms**.
This leads to **wasted time, poor productivity, and inefficient study management**.
There is a lack of **intelligent search and AI-based personalization** (auto-tagging, recommendations).
Existing e-learning platforms focus only on **content delivery**, not resource management.
Book resale/exchange platforms are **not integrated with academic study tools**.
No unified system exists that combines **study tools + resource sharing + marketplace**.
Limited availability of **secure and student-focused transaction systems** in education platforms.

SDG Alignment:

Aligns with **United Nations Sustainable Development Goal 4 (Quality Education)**
Aims to ensure **accessible, affordable, and inclusive learning resources** for students
Promotes **efficient learning, collaboration, and equal opportunities in education**

OBJECTIVES

General Objectives

Develop a **unified platform** for study management, resource sharing, and book exchange
Enable **AI-powered features** such as auto-tagging and personalized recommendations
Provide **secure and reliable transactions** for book exchange
Improve **student productivity and collaboration** through integrated tools
Ensure **easy accessibility and affordability** of academic resources

SDG Alignment

Supports **United Nations Sustainable Development Goal 4 (Quality Education)**
Promotes **inclusive and equitable access** to learning materials
Enhances **learning efficiency through technology and collaboration**
Encourages **affordable education via resource sharing and reuse**

OBJECTIVES

SMART Objectives

- **Specific:**
  Develop a **fully functional AI-powered study management system** integrating resource sharing, book marketplace, and collaboration tools
- **Measurable:**
  - ❑ Achieve **≥80% accuracy** in AI-based recommendations and tagging
  - ❑ Ensure **≥95% success rate** in secure transactions
  - ❑ Maintain **100% role-based access control** for users
- **Achievable:**
  Use technologies like **React, Node.js, MongoDB, and AI models (TensorFlow)** to build a scalable and feasible system
- **Relevant:**
  Addresses the **lack of integrated academic platforms** and improves **resource accessibility and learning outcomes**
- **Time-bound:**
  Complete development and deployment within **6–8 months**, including testing and optimization phases

LITERATURE SURVEY

Ref. No.	Title / System	Author(s) / Year	Pros	Limitation	VedShare Solution
1	BookSwap – Second-hand Book Buying and Selling Web Application	**Rishabh Singh, Vibhor Jain, Rhythm Yadav, Ujjawal Jain, Preeti Gupta** (2022) (Peer-reviewed Journal)	Reduces book costs, secure transactions, user ratings improve trust.	Limited scope — only book exchange, no study management.	Integrates AI-driven study tools and academic collaboration features.
2	Challenges & Coping Strategies among Distance Education Learners	**Moses Segbenya & Francis Aning Anokye** (2022) (PubMed Central)	Identifies financial burdens, stresses timely access to resources.	Focuses on challenges, not solutions.	Provides AI-powered personalized learning resources and solutions.
3	Intelligent E-commerce Platform for Affordable Student Book Exchange & Donation	**R. Aditya Padwal; Mohit Punde; Ankita Ghadge; Bharti Sahu** (2023).	Streamlines second-hand exchange, secure payments.	Only commerce-based productivity and less academic tools.	Combines academic marketplace with productivity and learning tools.
4	Online Book Store Management System: Revolutionizing Literature Retail in the Digital Age	**S. Rathanasabapathy, V. Bhuvaneswari, K. Monisha & B. Vaidianathan** (2025) (ResearchGate)	Enhances book retail via browsing, purchase, and digital sustainability.	Focuses mainly on retail, less on academic resource sharing.	Unifies academic resources, book exchange, and study tools in one platform.
5	Digital Transformation of Academic Libraries in India: Exploring ICT Integration	**Awanish Rajput & Praveen Kumar Pandey** (2024).	Improves knowledge sharing, digital repositories, efficiency in access.	Library-focused, less on student-to-student interaction.	Enables peer-to-peer exchange, AI-based personalization, and resource sharing.
6	Adoption & Effectiveness of Digital Collaboration Platforms: Indian Students' Perspective	**Archana Singh, Sarika Sharma, Manisha Paliwal** (2021 / published ahead-of-print 2020) (ResearchGate)	Interactive, cost-effective, enhances communication & learning.	Generic collaboration, not tailored for academic marketplace.	Tailors collaboration tools for academic needs with smart recommendations.

THE TIMES OF INDIA

Increase in cost of books hits parents

Kritika Tiwary / Mar 19, 2023, 08:16 IST

City Ranchi

Hindustan Times

New school books under NEP to cost 48% more

KANPUR Adding to the financial burden of parents, school books to be taught as a part of the new education policy from Classes 1st to 8th would get dearer by 48%

THE ECONOMIC TIMES | News

The cost of raising a child in India: School costs ₹30 lakh, college a crore

As per *ET Online* research, the overall expenditure of schooling a child in India in a **private school** from age 3 to age 17 is a whopping Rs 30 lakh.

THE TIMES OF INDIA

Costly books a major concern for parents

Venu Lanka / TNN / Updated: Jun 12, 2023, 11:13 IST

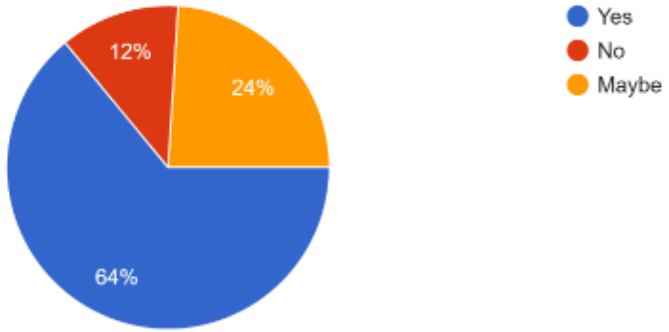
Skyrocketing price of books and other stationery items have become a major concern for parents ahead of reopening of schools from Monday.

mint

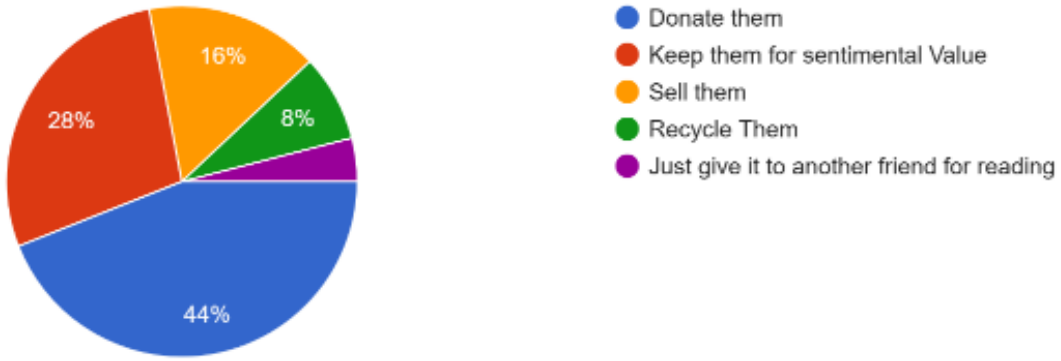
Cost of education in India: Exploring expenses and inflation rates

Jash Kriplani | 5 min read | 09 Jul 2023, 09:23 PM IST

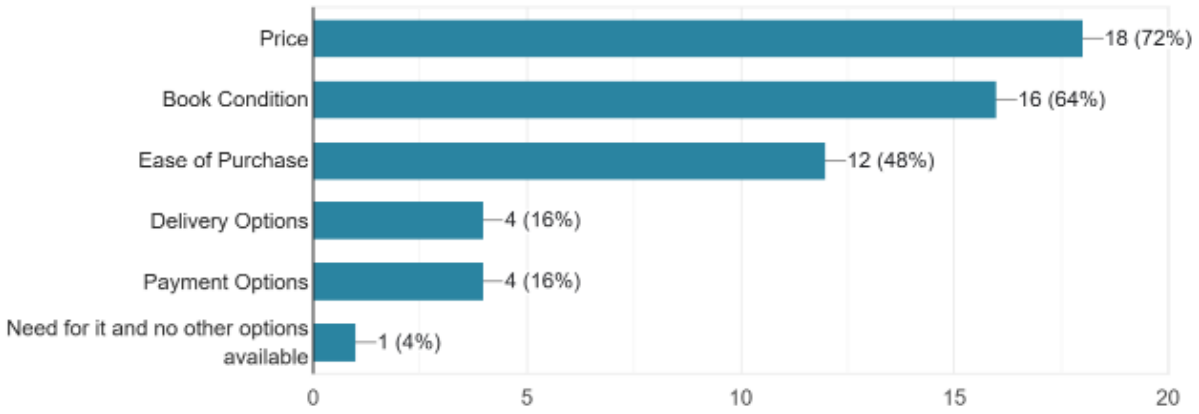
Would you consider buying second-hand books online when the need arises?



What do you usually do with your old books?



What factors would influence your decision to buy second-hand books online ? (Check all that apply)



LITERATURE SURVEY

Subsequent Research after Key Papers

1. Book Exchange & E-Commerce

"BookSwap – Second-hand Book Buying & Selling" (2022) → extended by **Wang et al. (2022)** on strategic decisions in online used-bookstores and **Ghose (2006)** on market impact of used books.

Gap: Transaction-focused, limited academic scope.

VedShare: Adds study tools, collaboration, and academic utility.

2. Digital Libraries & ICT Integration

"Digital Transformation of Academic Libraries in India: Exploring ICT Integration" (Rajput & Pandey, 2024) → followed by studies on adoption, satisfaction, and comparative policies (2024–25).

Gap: Library/institution focus, less student interaction.

VedShare: Enables peer-to-peer exchange, AI personalization, and student-driven resources.

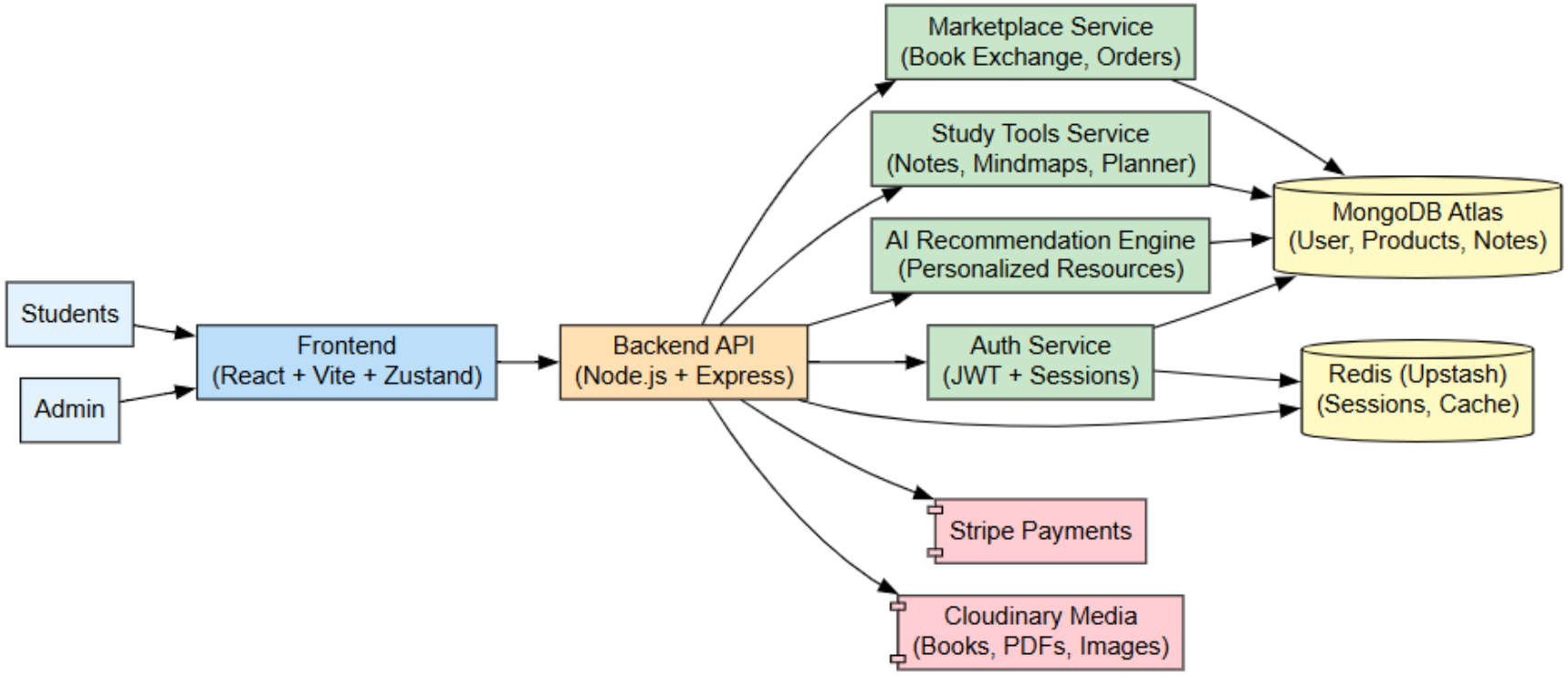
3. Digital Collaboration Tools

"Adoption & Effectiveness of Digital Collaboration Platforms: Indian Students' Perspective" (Singh et al., 2021) → later refined with personalization and engagement metrics (2023–25).

Gap: Generic platforms, not tailored for academics or resource exchange.

VedShare: Academic-focused collaboration with smart recommendations and integrated marketplace.

BLOCK DIAGRAM



SOFTWARE REQUIRENMENT

Backend Requirements

Software / Tool	Purpose	Use in Project	Cost
Node.js + Express	Web server & API routing	Backend APIs and routing	Free
MongoDB (Atlas)	NoSQL database	Store users & transactions	Free/Paid
Mongoose	Schema-based DB access	Define models/schemas	Free
bcryptjs	Password hashing	Secure user passwords	Free
jsonwebtoken	Token authentication	Manage access & refresh tokens	Free
Upstash Redis	Session persistence	Store refresh tokens	Free/Paid
Stripe	Payment processing	Handle transactions	Paid
Cloudinary	Media storage	Store PDFs, videos, images	Free/Paid
cookie-parser	Session management	Manage cookies securely	Free
dotenv	Env variable management	Store API keys/credentials	Free

SOFTWARE REQUIRENMENT

Frontend & Other Software Requirements

Software / Tool	Purpose	Use in Project	Cost
React + Vite	Frontend framework	Build UI	Free
axios	API communication	Connect frontend & backend	Free
react-router-dom	Client routing	Page navigation	Free
framer-motion	UI animations	Smooth transitions/UX	Free
react-hot-toast	Notifications	Alerts & status messages	Free
Cloud Infrastructure	Platform deployment	Deploy backend & frontend	Paid
Servers (Cloud/Ded.)	Hosting services	Run backend & DB	Paid
Cloud Storage (Cloudinary/Other)	Media storage	Store books & resources	Free/Paid

MARKET DEMAND ANALYSIS

Factor	Analysis	Source / Reference
Target Audience	Students (college & distance learners), Academic Institutions, Libraries	Online Survey.
Market Size	Global EdTech Market projected to reach **$404 Billion by 2026**, growing at a CAGR of ~16%	HolonIQ, 2023 EdTech Market Report
Current Solutions	Chegg, Coursera, BookSwap, OLX Books – focus on either content delivery or resale, not both	–
Unmet Need	Lack of an **integrated AI-powered study management + secure book marketplace** platform	Research Gap Identified from Literature Review

USP (**Unique Selling Proposition**) & SPECIALITY

1. What makes VedShare different?

1. Unlike existing e-learning or resale platforms, VedShare unifies study management + book marketplace + AI-driven personalization in a single secure system.
2. It not only enables affordable access to resources, but also improves learning efficiency through collaboration and smart recommendations.

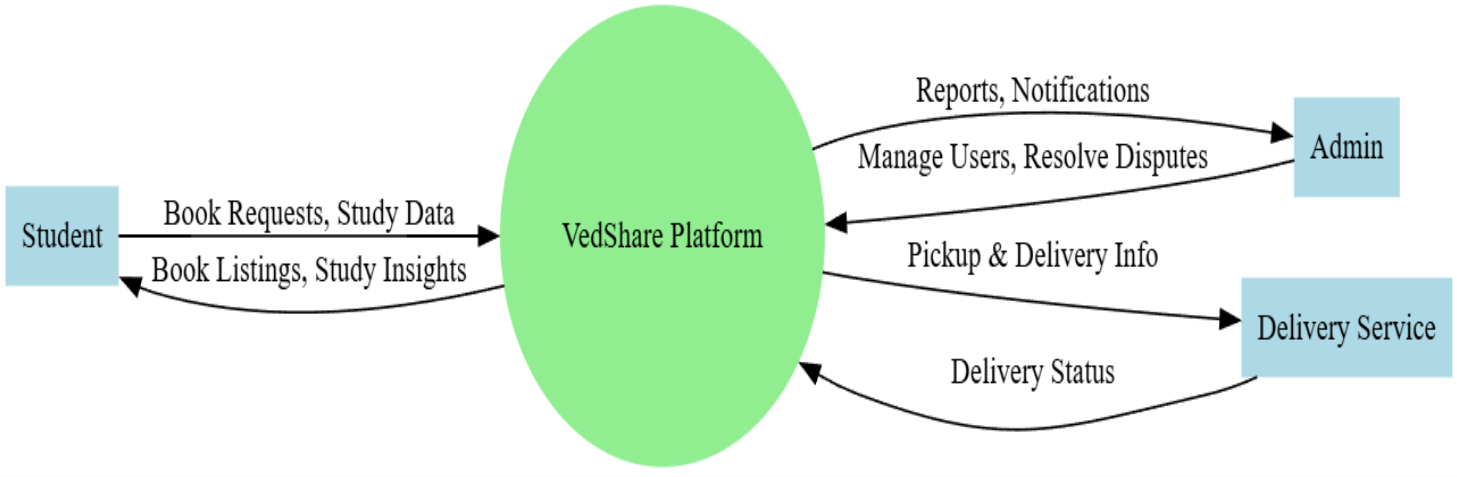
2. Key Unique Features

1. Integrated Platform – Combines e-learning tools, second-hand book exchange, and academic resource sharing under one roof.
2. AI-Powered Personalization – Smart recommendations for notes, books, and learning materials tailored to each student's study patterns.
3. Secure & Collaborative – Role-based dashboards, encrypted payments (Stripe), and peer-to-peer collaboration tools (notes, mindmaps, study groups).

Functional Design

Functional Design (Use Case):

- Book listing & search
- Secure second-hand book exchange
- Study management dashboard
- AI-based material recommendations
- Real-time collaboration (notes, discussions)
- Ratings, reviews, encrypted transactions

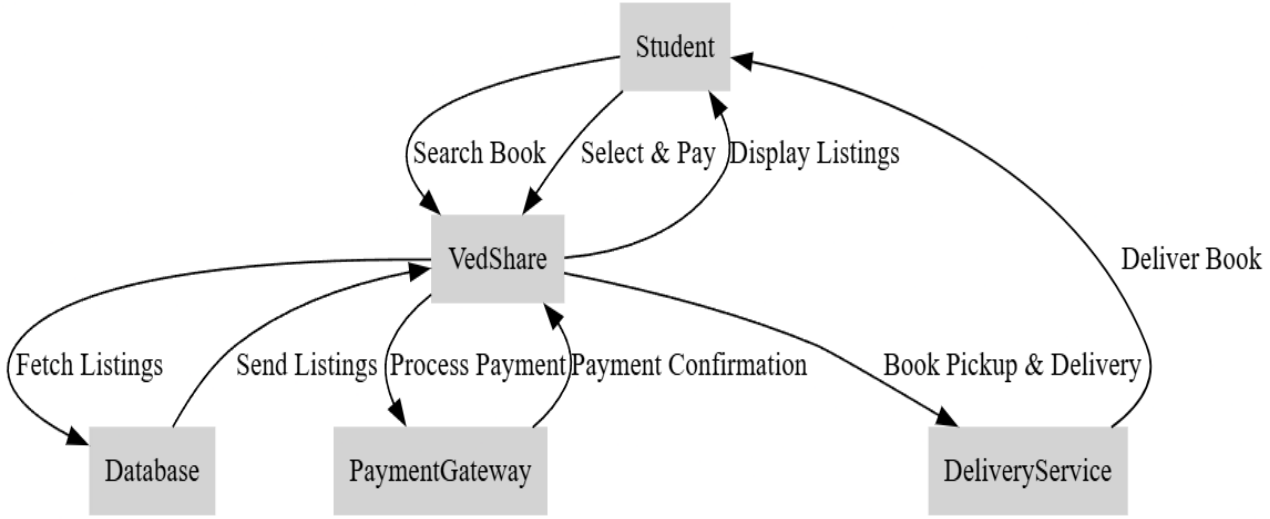


Functional Design

Behavioural Design

Behavioural Design (Interactions):

- User searches → Fetch from DB → Display listings
- Buyer requests book → Secure payment (Stripe) → Seller ships → Delivery tracked
- Students collaborate → Notes shared in real-time
- AI monitors usage → Suggests relevant books/resources

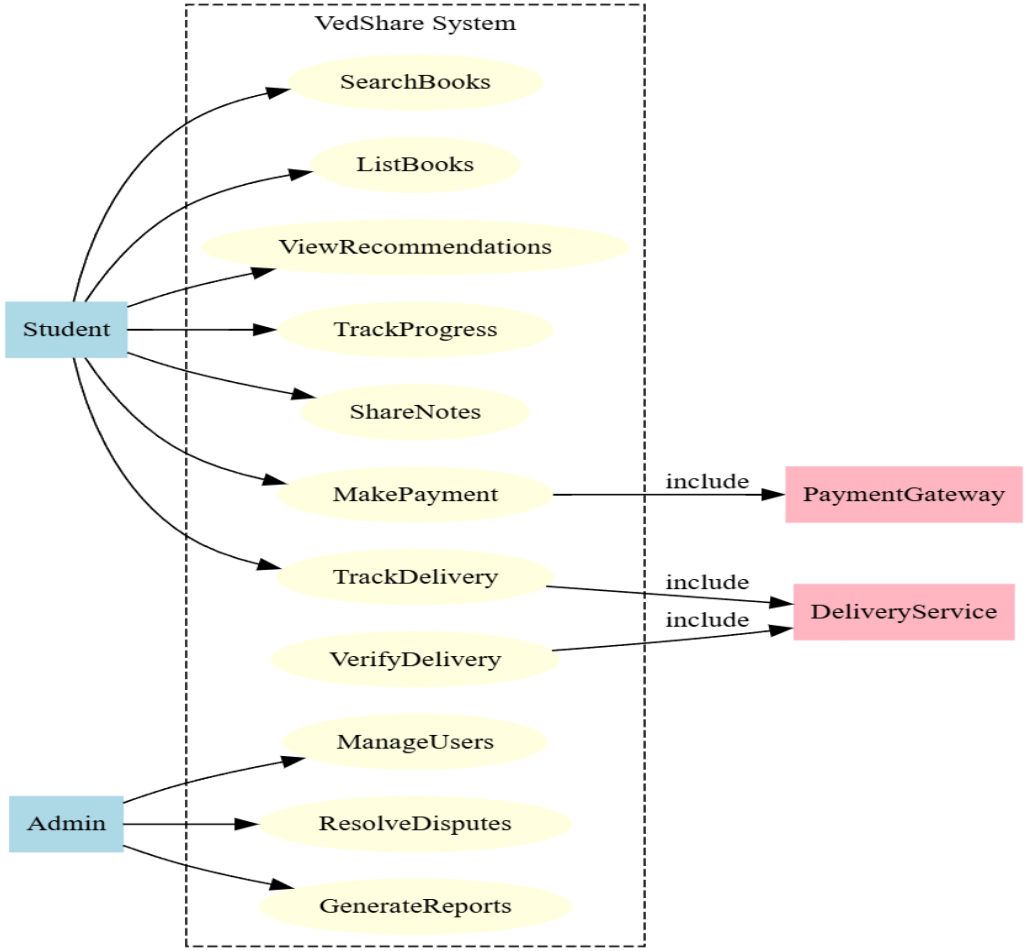


Behavioural Design

Structural Design

Structural Design (Components):

Frontend: React.js / Vite
Backend: Node.js + Express.js
Database: MongoDB Atlas
Auth: JWT
AI Engine: Python + TensorFlow
Media & Payments: Cloudinary, Stripe



Structural Design

Designing Methodology & Implementation Steps

Approach: Agile (iterative sprints)

Steps/Phases:

- Requirement gathering & feasibility study
- Architecture & database design
- Backend API development (Node.js + Express)
- Frontend development (React + Vite)
- Integration of Cloudinary & Stripe
- AI recommendation engine setup
- Testing & debugging
- Deployment (Vercel / Netlify / Heroku)

Tables of Comparative Results

Feature / Criteria	Traditional E-Learning	Book Exchange Platforms	VedShare
Resource Management	Limited	Not Available	Integrated System
Book Marketplace	Not Available	Available	Integrated
AI Recommendations	Minimal/None	None	
Personalization	Low	None	High
Secure Transactions	Not Focused	Basic	
Unified Platform	No	No	Yes
Accessibility & Affordability	Moderate	Moderate	High

Discussion

Results Interpretation:

Solves **resource fragmentation** by providing a **centralized platform**
Integrates **study tools + book marketplace + resource sharing** in one system
Uses **AI-based tagging & recommendations** to improve resource discovery
Reduces **manual effort and time wastage** in searching materials
Enables **secure peer-to-peer transactions** for book exchange
Improves **student collaboration** through notes, discussions, and sharing features
Enhances **learning efficiency and productivity**
Overcomes limitations of existing platforms (which focus only on content delivery)

SDG Alignment:
Supports **United Nations Sustainable Development Goal 4 (Quality Education)**
Promotes **accessible and affordable education**
Encourages **inclusive and collaborative learning**
Helps improve **overall academic performance and equal opportunities**

Conclusion

a. Technical Impact

VedShare demonstrates the successful integration of modern web technologies, cloud-based databases, and AI-driven modules into a unified academic resource platform. The system proves to be technically feasible, secure, and scalable, offering seamless management of study materials through services like the marketplace, study dashboard, and AI-powered tagging and recommendations. Its modular architecture ensures extensibility, making it adaptable for future enhancements.

b. Societal Impact

VedShare addresses the critical challenges faced by students by reducing resource fragmentation and promoting affordability through book exchange and resource sharing. The platform encourages collaboration and equal access to quality study materials, thus contributing to more inclusive and efficient learning experiences. By empowering students with intelligent tools for managing and discovering resources, VedShare has the potential to improve academic productivity and reduce educational disparities

Future Scope

1. VedShare can be enhanced further with **advanced AI recommendations and tagging** that suggest not only books but also **notes, question banks, and research papers**.

2. It can also support **peer learning together in real-time**, enabling students to collaborate and grow collectively.

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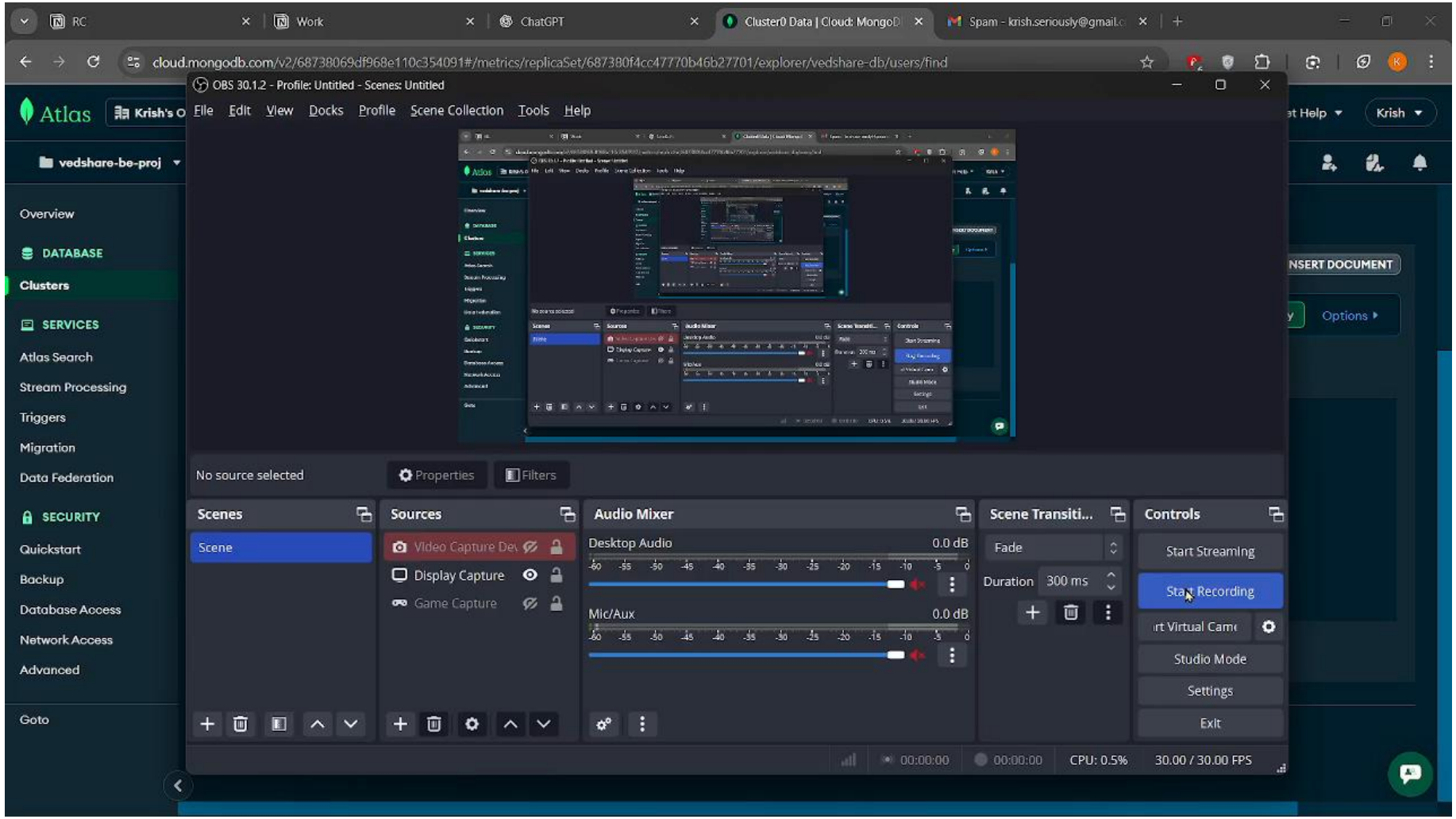
Short Video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Team & Guide Information

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